Can we find delivery types without labels?

K-means clustering, on real Hawk-Eye data

The cricket question

— Hawk-Eye prints speed, where it pitched, where it passed the stumps.

— Could you tell fast bowler from spinner on those numbers alone?

— Left arm from right arm? Or do the balls group some other way?

The data

— 140,233 IPL deliveries, 5 Hawk-Eye measurements per ball.

— Tracker failures and impossible readings dropped (~6%).

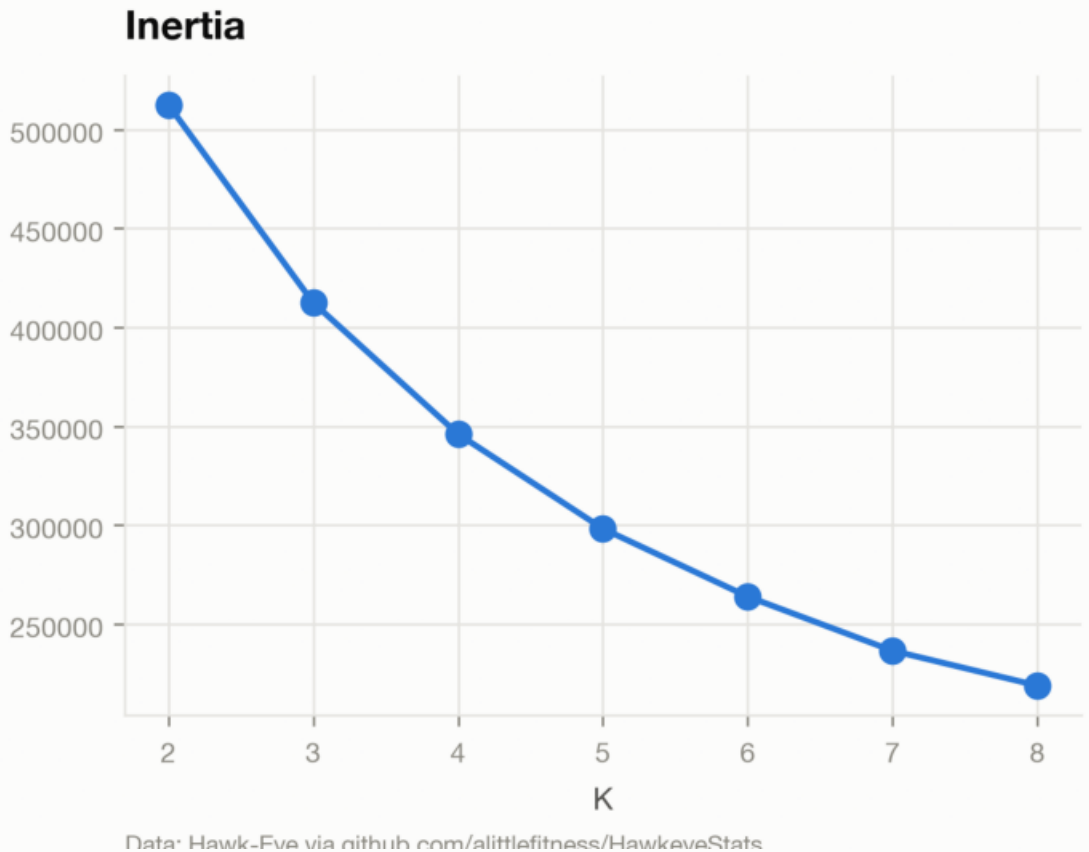
— bowler + arm/style held out — used only to interpret clusters.

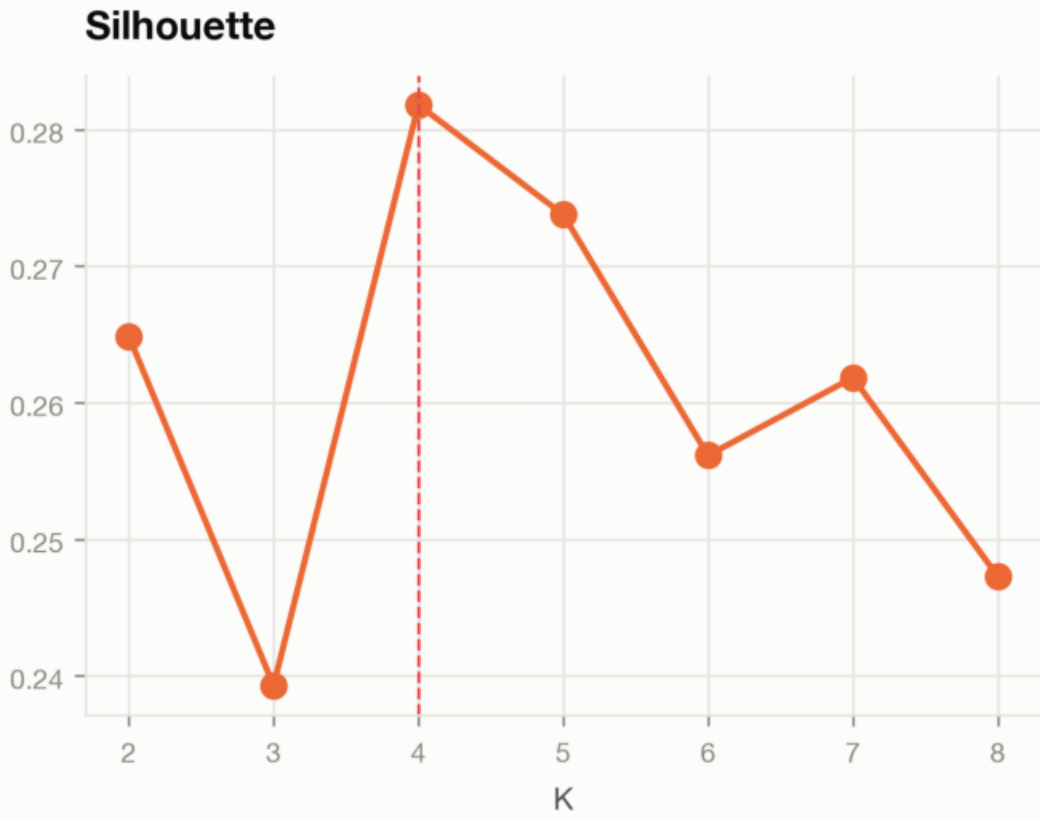
— Source: github.com/alittlefitness/HawkeyeStats (Hawk-Eye via the ICC).

The naive answer

- The scorecard already labels every bowler: arm + style.
- RIGHT_FAST_SEAM, LEFT_ORTHODOX, RIGHT_LEG_SPIN, ...
- Coarsest split, pace vs spin, must be recovered before anything else counts.
- Caveat: the 'SEAM' fallback style has an unreliable arm flag.

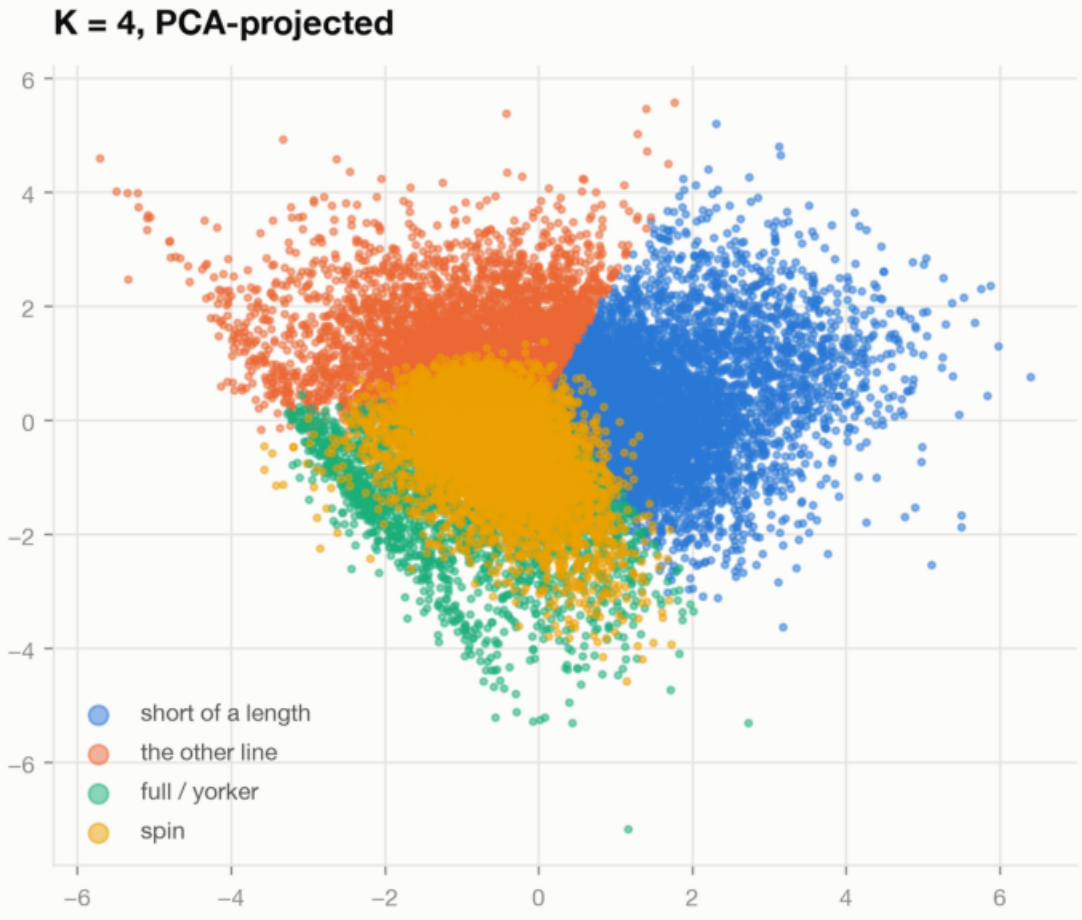
Choosing K: elbow + silhouette





Silhouette peaks at K=4

K = 4, projected to 2D



What the clusters found

archetype	share	kph	length (m)	spin share	left-hand batter
spin	32%	91	5.3	80%	29%
the other line	20%	119	5.8	25%	74%
short of a length	32%	132	9.0	0%	27%
full / yorker	16%	133	3.4	3%	24%

ARI vs pace/spin 0.25; vs bowling arm 0.00

What it gets wrong

- 'The other line' is left-handed batters: X coordinates are pitch-fixed.
- Arm is invisible: release position isn't in these five columns.
- Clusters are deliveries, not bowlers; every seamer bowls full and short.
- Standardising gives line and height the same vote as speed.

Next: compressing these features with PCA

cricketml